ELSEVIER

Earth and Planetary Science Letters 153 (1997) 157–169

EPSL

Paleomagnetism of the Paleoproterozoic Gowganda and Lorrain formations, Ontario: low paleolatitude for Huronian glaciation

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Received 18 June 1997; revised 24 September 1997; accepted 24 September 1997

Abstract

To further our understanding of the paleolatitudes of Precambrian glaciations, a paleomagnetic study has been conducted on the glaciomarine Coleman Member of the Gowganda Formation and the conformably overlying deltaic Firstbrook Member and Lorrain Formation (Huronian Supergroup, ~ 2.4–2.3 Ga) in Ontario, Canada. Many of the rocks, notably the gray and grayish red argillic facies, display unstable magnetization or overprints carried by magnetite and/or hematite that give negative fold tests. Some red sandstones, however, have stable high-temperature (~ 690°C) components marked by very shallow northerly or southwesterly directions. Five sites in pale red sandstone from the Coleman Member near Elliot Lake give a dip-corrected A direction of $D = 216.3°$, $I = 5.5°$ ($\alpha_{95} = 6.7°$, $n = 30$), and 17 sites in Lorrain red sandstone near Desbarats give a dip-corrected A direction of $D = 5.4°$, $I = 5.5°$ ($\alpha_{95} = 5.9°$, $n = 75$). Unfavorable structural attitudes rendered fold tests inconclusive. These very shallow directions may be ascribed to early chemical remanent magnetization (CRM) because (a) some specimens display polarity reversals, (b) such directions are unknown, either as original or as overprint components, in post-Huronian rocks of the region, and (c) a prior site in Coleman red beds gave a high-temperature, shallow NNW direction that is not significantly different from Lorrain A and which a positive conglomerate test suggested is not an overprint. The very shallow directions are near primary directions for the slightly older (2.45 Ga) Matachewan dyke swarm in the region. Lorrain argillic rocks at five sites give a normal component B ($T = 670°C$, $D = 18.6°$, $I = 59.6°$, $\alpha_{95} = 6.1°$, $n = 16$, dip corrected) and a reverse component E ($T = 500°C$, $D = 190.9°$, $I = -60.9°$, $\alpha_{95} = 7.3°$, $n = 14$, in situ), which are close to a prefolding direction carried by magnetite previously identified in Coleman gray argillites (Morris A). However, a primary origin for Morris A is dubious because of the widespread occurrence of overprints carried by magnetite in Coleman gray argillites and because the fold test employed folds formed during the Penokean Orogeny at 1900–1800 Ma. Our results suggest that Morris A/Lorrain B and Lorrain E may date from immediately before and just after Penokean folding, respectively. Hence Coleman-sandstone A and Lorrain A may be better estimates of original directions. Those components and Matachewan paleomagnetic data constrain Huronian paleolatitudes and suggest that Huronian glaciation occurred within 11° and possibly within 4° of the paleoequator. © 1997 Elsevier Science B.V.

Keywords: Precambrian; glaciation; paleomagnetism; Huronian; Gowganda Formation; Lorrain Formation

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PII S0012-821X(97)00181-7

1. Introduction

A major controversy in contemporary earth science concerns the paleolatitudes of Precambrian glaciations, with conflicting interpretations placing glaciation in near-equatorial paleolatitudes [1] or no closer than ~25° to the paleo-equator [2]. Recent paleomagnetic evidence suggests, however, that some Neoproterozoic glaciomarine successions indeed accumulated near the paleoequator: the Marinoan glaciation (~620 Ma) in South Australia at a paleolatitude of $2.7 \pm 3.7°$ [3] and Rapitan (~725 Ma) glacial episodes in northwestern Canada at paleolatitudes of $6 \pm 4°$ and $4 \pm 6°$ [4]. Paleolatitudes of Paleoproterozoic glaciations have been more elusive, but recently Evans et al. [5] obtained a paleolatitude of $11 \pm 5°$ for ~2.2-Ga volcanics overlying the glaciogenic Makganyene diamictite in South Africa, and arguing that volcanism closely followed glaciation they inferred a low paleolatitude of glaciation.

Paleomagnetic data for the glaciogenic Gowganda Formation of the ~2.4–2.3-Ga Huronian Supergroup in Ontario, Canada, require review because the most commonly cited study [6] predates the widespread use of orthogonal projections and principal component analysis and investigated only a drab argillic facies, and because of widespread overprinting of such Huronian sediments. Hence, a paleomagnetic study has been conducted on the glaciomarine Coleman Member and deltaic Firstbrook Member of the Gowganda Formation and the conformably overlying Lorrain Formation with the aim of determining the paleolatitude of glaciation. Our findings for hematitic sandstones from the Coleman Member and the Lorrain Formation, together with previous paleomagnetic data for the Gowganda Formation and for

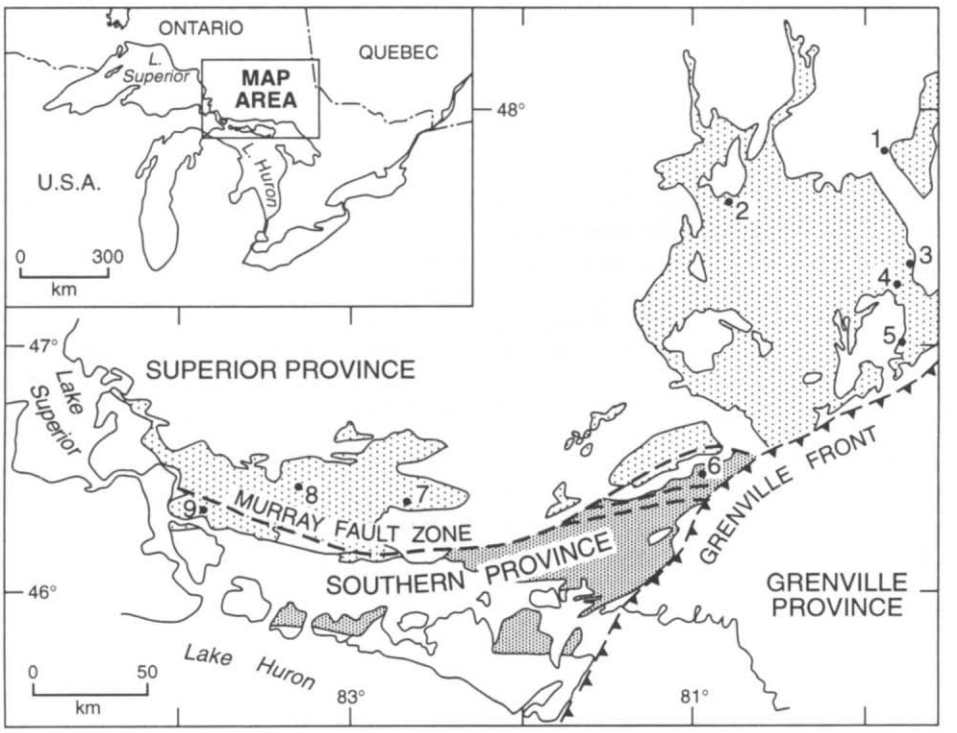


Fig. 1. Distribution of the Huronian Supergroup in the Southern Province, Ontario (adapted from [7]): thin, little deformed, unmetamorphosed succession (light stipple); thick, deformed, metamorphosed succession (dense stipple). *1*, Englehart. *2*, Gowganda. *3*, Cobalt. *4*, Latchford. *5*, Temagami. *6*, Sudbury. *7*, Elliot Lake. *8*, Cumming Lake. *9*, Desbarats.

mafic dyke swarms in the southern Superior Province, rule out high paleolatitudes and suggest low paleolatitudes for Huronian glaciation.

2. Huronian Supergroup

The Huronian Supergroup in the Southern Province of Ontario (Fig. 1) has among the world's oldest and best preserved glaciogenic successions [8,9]. The supergroup comprises four groups (Fig. 2), the three topmost each displaying tripartite cyclicity commencing with glaciogenic diamictite facies—the Ramsay Lake, Bruce and Gowganda formations—followed by an argillic unit then cross-bedded sandstones. The age of the glaciogenic successions is constrained by the ages of 2473–2446 Ma for the Matachewan mafic dyke swarm that intrudes basement rocks of the Superior Province [10] and 2450 Ma for coeval volcanics near the base of the Huronian Supergroup [9], and the age of 2217–2210 Ma for Nipissing Diabase that intrudes the supergroup [10].

The Gowganda Formation [11,12] is the youngest and most widespread of the glaciogenic successions and is 600–1700 m thick. The Coleman Member includes gray and black diamictite, gray and black argillite–dropstone facies locally displaying varve-like lamination, conglomerate, and pale to moderate red (5R 6/2–5R 5/4) and pale reddish brown (10R 5/4), well sorted, mostly medium-grained feldspathic sandstone (color designations from the Geological Society of America Rock-Color Chart, 1963). Selected sandstone samples averaged 0.57% Fe_2O_3 (range 0.40–0.73%) and 0.77% FeO (0.34–1.49%). Most of the iron is contained in scattered grains (5–100 μm) of detrital hematite, some with exsolution lamellae of ilmenite, irregular and platy grains of secondary hematite, and euhedral secondary magnetite (20–200 μm) partly altered to hematite.

In the Latchford–Cobalt area the Coleman Member is conformably overlain by the Firstbrook Member of non-glaciogenic deltaic origin [13]. The Firstbrook Member is up to 535 m thick and comprises grayish red (5R 4/2) argillic siltstone that passes upward into a sandstone-dominated facies. Selected samples of argillic siltstone averaged 3.16% Fe_2O_3 (2.51–3.76%) and 4.94% FeO (4.22–5.58%), most of the iron being contained in ultrafine matrix material.

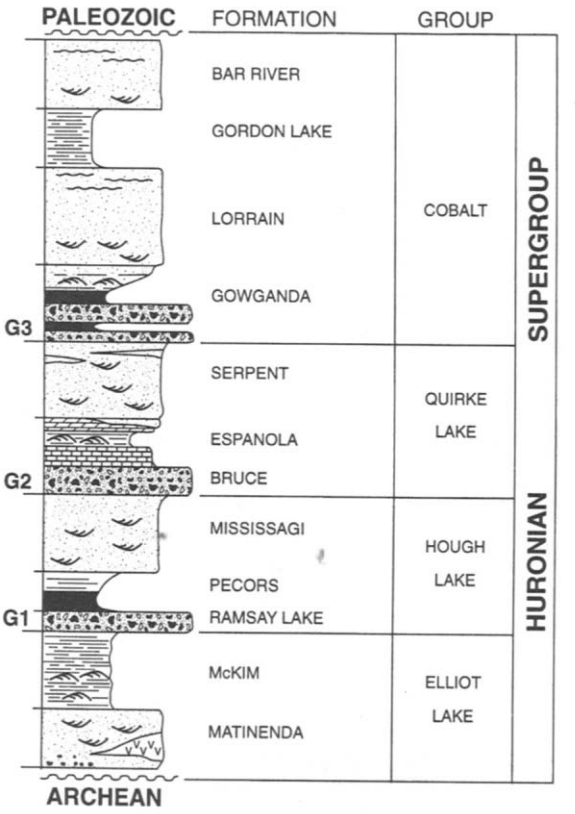


Fig. 2. Generalized stratigraphy of the Huronian Supergroup (maximum thickness ~ 12,000 m), showing the glaciogenic diamictite facies of the Ramsay Lake, Bruce and Gowganda formations (G1–G3). The diamictite facies of the Gowganda Formation (Coleman Member) is conformably overlain by a coarsening-upward deltaic unit (Firstbrook Member). Adapted from [9].

The deltaic Lorrain Formation comprises up to 2000 m of sandstone and conglomerate and conformably overlies the Gowganda Formation [13,14]. Our sampling of the Lorrain Formation was confined to the 120-m thick 'purple siltstone' member near the base of the formation in the Desbarats area. This member is a red bed facies of moderate red (5R 4/4), well sorted, fine-grained feldspathic sandstone in beds up to 2 m thick and local grayish red (5R 4/2) argillic siltstone and fine-grained sandstone in graded beds 2–10 cm thick. Selected sandstone samples averaged 2.65% Fe_2O_3 (1.60–3.63%) and 1.10% FeO (0.36–3.35%). Most of the iron is contained in

an intergranular ultrafine hematitic pigment. Scattered opaque grains (2–100 μm) in the sandstone include detrital hematite, some with exsolution lamellae of ilmenite, and irregular grains of secondary hematite; secondary pyrite and rare chalcopyrite also are present in the argillic rocks.

The Huronian rocks have experienced at least two distinct folding events [15–17]. An early event, which predated intrusion of Nipissing diabase sheets at 2219 Ma, produced northerly-trending folds in the northeast and south, notably in the area south of the Murray Fault Zone (Fig. 1) immediately west of Sudbury and near Lake Huron. A more important, post-Nipissing tectonic event produced widespread folding of the Huronian Supergroup, including easterly-trending open folds and southerly tilted fault blocks in the western belt to the north of the Murray Fault Zone and much tighter easterly- to northeasterly-trending folds south of the fault zone. This second deformation, which is regarded as coeval with the Penokean Orogeny at 1900–1800 Ma, comprised pre-1850 Ma and post-1850 Ma phases [16]. Episodic earth movements and faulting continued to the time of the Grenville Orogeny at ~ 1000 Ma.

3. Previous paleomagnetic studies of Huronian Supergroup

Paleomagnetic study of the Huronian Supergroup is made difficult by widespread overprinting related to diabase intrusions and regional deformation, and little work has been done since the early 1980s.

Symons [18] sampled gray and reddish tillite, graywacke, siltstone and arkose through the Coleman Member near Elliot Lake. A persistent direction ($D = 326°$, $I = 76°$, in situ), evidently carried by magnetite, was interpreted as early diagenetic and indicating a paleolatitude of 62° for glacial deposition.

Morris [6,19] sampled only gray varvelike argillites from the Coleman Member near Cobalt, Elliot Lake, Cumming Lake and other localities and from the probably equivalent Chibougamau Formation in Quebec. Initially Morris [19] found that the "limitation of remanence age by fold and conglomerate tests is not conclusive", but subsequently [6] he identified a prefolding A direction ($D = 29°$, $I = 52°$, dip corrected) and a postfolding B direction ($D = 336°$, $I = 68°$, in situ) each carried by magnetite. Morris B is near the direction of Symons [18], which Morris [6] reinterpreted as an overprint related to regional deformation. Such a postfolding diagenetic or Penokean metamorphic direction ($D = 327°$, $I = 78°$, in situ), evidently carried by magnetic, was confirmed in Coleman sediments near Cobalt [20]. However, a primary origin for Morris A [6] has not been demonstrated satisfactorily. The origin of the magnetite carriers in his argillite samples is unclear: an illustrated grain 5 μm across, described as comprising a euhedral magnetite overgrowth on detrital magnetite [6], has the appearance of an authigenic magnetite grain with an altered rim. Moreover, Morris's [6] fold test employed easterly-trending folds, including those in the Elliot Lake area, that are of Penokean age [17] and hence his fold test only loosely constrains the timing of his A direction.

The Thessalon basalts, which occur in the Matinenda Formation near the base of the Huronian Supergroup (Fig. 2) at Thessalon ~ 30 km east of Desbarats, carry a remanence in titanomagnetite ($D = 304°$, $I = 54°$, in situ) that Symons and O'Leary [21] interpreted as original and indicating extrusion at a paleolatitude of 35°. Subsequently, Stupavsky and Symons [22] found that the coeval Dollyberry Lake basalts ~ 15 km north of Elliot Lake display two postfolding components: component A residing in both coarse- and fine-grained magnetite and in hematite ($D = 316–329°$, $I = 63–72°$, in situ) and evidently acquired during the Penokean Orogeny, and a later component B carried by fine-grained magnetite ($D = 338°$, $I = -37°$, in situ). They found that the Thessalon basalts retain the A component and retracted the finding [21] of extrusion at ~ 35° paleolatitude.

Roy and Lapointe [23] studied grayish red argillic siltstone of the Firstbrook Member near Latchford and Cobalt and identified a high-temperature component (Group B: $T = 650°C$, $D = 340°$, $I = 56°$, in situ) which they suggested was acquired shortly after deposition. In addition, they found a direction for one site in the Coleman Member near Cobalt (Group A: $T = 650°C$, $D = 10°$, $I = 34°$, dip corrected), which they suggested also is primary. Stupavsky et al. [24] studied the Firstbrook argillic siltstone near Cobalt and identified two components ($T = 625°C$,

$D = 332°$, $I = 60°$, in situ; $T = 625°C$, $D = 279°$, $I = -52°$, in situ) which they viewed as primary despite the directions being highly oblique, and equated the normal component with Roy and Lapointe B [23]. The directions of these Firstbrook normal components, however, are close to the widespread postfolding Penokean direction identified for the Coleman Member [6,20] and lower Huronian volcanics [22].

More recently, Buchan et al. [25] found that gray sediments of the Coleman Member "are not magnetically stable except where altered chemically by a nearby intrusion", and concluded that "the grey units are of no use in establishing a primary Coleman paleopole". They noted, however, that potential exists for obtaining primary directions from Coleman red beds if they are far removed from the influence of intrusions. They identified a site in Coleman red beds near Englehart for which three specimens gave a stable, shallow NNW-down direction ($T = 620$–$700°C$, hematite carrier, D Å 327°, I Å 34°, in situ) that is unlike directions observed in either a nearby Nipissing sill or crosscutting dyke. A positive conglomerate test suggested to Buchan et al. [25] that this stable shallow direction is not an overprint.

In conclusion, overprints related to Nipissing intrusions [20,23,25] and a postfolding diagenetic or Penokean metamorphic direction [6,20,22] have been identified in the Huronian Supergroup throughout the Southern Province. Gowganda remanence directions of uncertain origin include Morris A in Coleman gray argillite [6] and single sites in Coleman hematitic sediments suggesting shallow northwesterly and northerly directions [23,25]. Hence, available data have not established an early remanence direction for the Gowganda Formation and further paleomagnetic work, particularly on Huronian red bed facies, appears warranted.

4. Paleomagnetism of the Gowganda and Lorrain formations

4.1. Sampling and techniques

Samples were taken from Coleman diamictite, argillite and sandstone near Gowganda, Cobalt, Latchford, Temagami, Elliot Lake and Cumming Lake, Firstbrook argillic siltstone near Cobalt and Latchford, and the Lorrain 'purple siltstone' member near Desbarats (Fig. 1). Samples were taken at some of Morris's [6] sites in Coleman argillite near Elliot Lake and Cumming Lake and at all of Roy and Lapointe's [23] Firstbrook sites. The Nipissing Diabase also was sampled between Cobalt and Desbarats. Huronian strata in this northern zone of the Southern Province usually have low dips and the regional metamorphic grade does not exceed lower greenschist [7], although diabase intrusions have altered nearby strata in many places. Most samples were drilled in the field and oriented by sun and magnetic compasses.

Results for reconnaissance sampling in 1995 confirmed that black, gray and grayish red argillic facies and drab diamictite matrix display unstable magnetization or overprints attributable to intrusions or folding. However, this initial work also confirmed that stable magnetization may be shown by some red sandstones in the Coleman Member and Lorrain Formation, and those lithologies were further sampled in 1996 and 1997.

All samples were subjected to stepwise thermal demagnetization and some also to stepwise alternating-field demagnetization. Thermal demagnetization was the most effective method of resolving the magnetic components in the sedimentary rocks. Selected specimens also were subjected to 3-axis isothermal remanence analysis following the method of Lowrie [26]. The Lorrain specimens show small but distinctive remanence decay just below 580°C which is interpreted as indicating the presence of magnetite. This fraction of the magnetite has coercivity $H_c >$ 100 mT and therefore is very fine grained, probably < 1 μm. Components of magnetization were isolated using a WINDOWS version of LINEFIND [27]. In this procedure, linear segments are fitted to data points in plots of orthogonal projections, the data points being weighted according to the inverse of their measured variances.

In all, 574 specimens from 86 sites were demagnetized but little more than half of this material provided useable results. Here we present results for seven sites in gray argillic diamictite and 15 sites in pale red sandstone of the Coleman Member, 11 sites in grayish red argillic siltstone of the Firstbrook Member, and 17 sites in the Lorrain 'purple siltstone' member.

4.2. Results

Intensities of natural remanent magnetization (NRM) and low-field susceptibility are bimodally distributed for Coleman sandstone, some sites displaying significantly stronger remanence and higher susceptibility than others. The more strongly magnetized Coleman sandstone sites had NRM intensities of 100–300 mA m^{-1} and susceptibilities of 0.9–1.2 $\times 10^{-2}$ SI, whereas the more weakly magnetized sites had values of < 1 to 50 mA m^{-1} and 0.6–1.2 $\times 10^{-3}$ SI. Most of the Coleman sandstone samples yielding low inclinations (see below) possessed the higher NRM intensities and higher susceptibilities. Although discrete unblocking temperatures near 690°C indicate hematite to be the remanence carrier, the high susceptibilities suggest that the sandstone contains some magnetite. Hysteresis measurements do indeed disclose a low coercivity fraction with coercivity $H_c < 80$ mT. However, magnetite is not apparent in the thermal demagnetization of Coleman sandstone. Coleman NRM directions show a pronounced grouping directed northward and steeply down, evidently reflecting a large proportion of present or recent field components.

The Lorrain sandstone possesses more uniform properties, with mean NRM intensity and susceptibility of 1–15 mA m^{-1} and 0.5–5.0 $\times 10^{-4}$ SI. Lorrain NRM directions formed a loose group spread over the northwest- to northeast-down octant.

Most Firstbrook sites gave NRM intensities of 23.9 ± 12.3 mA m^{-1} and susceptibilities of $1.4 \pm 0.3 \times 10^{-4}$ SI, whereas several more strongly magnetic sites gave mean NRM intensities of 19.6 ± 5.3 mA m^{-1} and susceptibilities of $5.0 \pm 1.8 \times 10^{-3}$ SI. NRM directions were clustered near the present-recent magnetic field direction, with some scattered to the southeast and down.

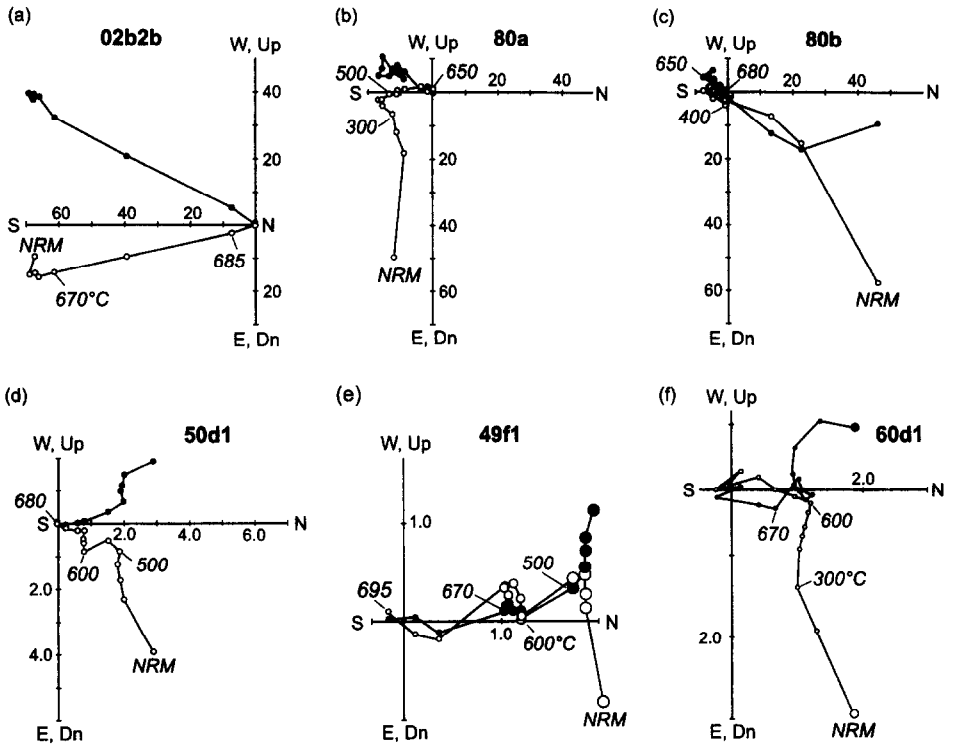


Fig. 3. Orthogonal projections showing examples of behavior on stepwise thermal demagnetization, from NRM to ≥ 680°C, of specimens of (a–c) pale red sandstone from the Coleman Member near Elliot Lake, and (d–f) red sandstone from the Lorrain Formation near Desbarats. Closed circles represent magnetization vectors projected onto the horizontal plane and open circles represent projections onto the vertical plane. Circle size indicates the circular standard error of each measurement. Units are mA m^{-1} ($= 1\ \mu G = 10^{-6}$ emu/cc). *Dn*, Down. Specimen numbers refer to site/sample/specimen.

Coleman gray argillic diamictite from near Temagami, Elliot Lake and Cumming Lake had NRM intensities of 2.07 ± 1.85 mA m^{-1} and susceptibilities of $2.0 \pm 0.7 \times 10^{-4}$ SI. NRM directions were scattered over the lower hemisphere with a few on the upper hemisphere.

With stepwise thermal demagnetization (Fig. 3), directions in many specimens changed rapidly up to 400°C with the removal of viscous magnetization. The Coleman sandstone possesses a stable component with a high unblocking temperature (~ 690°C), as does the Coleman gray diamictite ($T = 650$°C).

Table 1
Paleomagnetic results for the Coleman and Firstbrook members of the Gowganda Formation and for the Lorrain Formation, Ontario

Formation and Component	T (°C)	$N(n)$	D_h (°)	I_h (°)	k	α_{95} (°)	D_b (°)	I_b (°)	k	α_{95} (°)
Lorrain Formation (red fine-grained sandstone and minor grayish red argillic siltstone from the 'purple siltstone' member)										
F (13)	450	13(58)	250.9	57.5	40.8	3.0	254.7	47.9	18.0	4.5[a]
Pole (in situ): Lat. = 15.6°N, Long. = 225.5°E, dp = 3.2°, dm = 4.3°.										
E (12)	500	5(14)	190.9	−60.9	30.3	7.3	181.7	−50.6	29.6	7.4
Pole (in situ): Lat. = 81.5°N, Long. = 34.4°E, dp = 8.5°, dm = 11.1°.										
D (11)	580	6(29)	335.8	−26.1	114	2.5	340.0	−27.6	41.0	4.2[a]
Pole (in situ): Lat. = 26.0°N, Long. = 122.4°E, dp = 1.4°, dm = 2.7°.										
C (10)	650	9(20)	239.6	61.1	57.6	4.3	248.8	53.7	20.2	7.4
Pole (in situ): Lat. = 13.1°N, Long. = 235.0°E, dp = 5.0°, dm = 6.5°.										
B (9)	670	5(16)	39.3	64.7	37.0	6.1	18.6	59.6	37.3	6.1
Pole (in situ): Lat. = 63.2°N, Long. = 351.1°E, dp = 7.8°, dm = 9.8°.										
Pole (dip corrected): Lat = 75.3°N, Long. = 23.4°E, dp = 6.9°, dm = 9.1°.										
A (8)	690	17(75)	6.6	9.8	9.0	5.8	5.4	5.5	8.7	5.9
Pole (in situ): Lat. = 48.1°S, Long. = 266.2°E, dp = 3.0°, dm = 5.9°.										
Pole (dip corrected): Lat. = 46.1°S, Long. = 268.3°E, dp = 3.0°, dm = 5.9°.										
Firstbrook Member (grayish red argillic siltstone), Gowganda Formation										
C (7)	500	7(29)	99.9	85.4	38.0	4.4	51.4	83.3	21.6	5.9[a]
Pole (in situ): Lat. = 45.0°N, Long. = 293.0°E, dp = 8.6°, dm = 8.7°.										
B (6)	680	3(17)	278.3	−55.3	36.6	6.0	283.3	−68.6	15.0	9.5[a]
Pole (in situ): Lat. = 20.5°S, Long. = 159.1°E, dp = 6.1°, dm = 8.5°.										
A (5)	680	8(50)	334.1	63.0	47.8	4.4	339.1	60.7	19.3	4.7[a]
Pole (in situ): Lat. = 71.8°N, Long. = 190.7°E, dp = 5.4°, dm = 6.9°.										
Coleman Member (gray argillic diamictite), Gowganda Formation										
B (4)	300	5(21)	348.6	71.6	31.3	5.8	318.9	41.1	3.0	22.5[a]
Pole (in situ): Lat. = 77.8°N, Long. = 246.0°E, dp = 8.9°, dm = 10.1°.										
A (3)	650	7(28)	153.1	70.4	2.9	21.2	271.9	69.6	13.0	8.5[b]
Pole (dip corrected): Lat = 36.5°N, Long. = 229.5°E, dp = 12.4°, dm = 14.5°.										
Coleman Member (pale red sandstone), Gowganda Formation										
B (2)	300	15(60)	348.0	83.9	16.7	4.6	6.8	84.1	13.5	5.2[a]
Pole (in situ): Lat. = 58.2°N, Long. = 272.7°E, dp = 8.8°, dm = 9.0°.										
A (1)	690	5(30)	215.9	1.2	15.8	6.8	216.3	5.5	16.3	6.7
Pole (in situ): Lat. = 33.4°S, Long. = 232.9°E, dp = 3.4°, dm = 6.8°.										
Pole (dip corrected): Lat. = 31.3°S, Long. = 233.7°E, dp = 3.4°, dm = 6.7°.										

$N(n)$, number of sites (specimens); D_h, I_h, mean declination and mean inclination with respect to present horizontal; D_b, I_b, mean declination and mean inclination with respect to bedding; k, precision parameter; α_{95}, semi-angle of cone of 95% confidence about mean direction; T, unblocking temperature; dp, dm, semi-major and semi-minor axes of polar error ellipse.

[a] Negative fold test at > 95% confidence.

[b] Positive fold test at > 95% confidence. All other fold tests inconclusive.

The Firstbrook argillic siltstone and the Lorrain sandstone have components with moderate (450–580°C) and high (650–690°C) unblocking temperatures. These results are consistent with magnetite and hematite being the principal carriers of remanence.

The 13 components identified are listed in Table 1: two components for Coleman sandstone, two for Coleman diamictite, three for the Firstbrook Formation, and six for the Lorrain Formation. All are ordered for each facies or formation with respect to unblocking temperature, and also are numbered from 1 to 13 for use in later plots. The results of McFadden [28] fold tests are indicated in the right-hand column. Table 1 also gives paleomagnetic pole positions for the components.

Coleman-sandstone B, Coleman-diamictite B and Firstbrook A–C have steep directions that yield negative fold tests. Coleman-diamictite A has a steep direction that gives a positive fold test.

Lorrain B–F have shallow to steep directions, whereas Lorrain A is very shallow. The northerly to northeasterly strike of strata around the nose of the west-northwesterly-plunging Thessalon Anticline near Desbarats [14], where the 'purple siltstone' member crops out, did not permit conclusive fold tests for all these components. Lorrain D and F yield negative fold tests. Lorrain C becomes more scattered on dip-correction but does not change significantly; however, its directional similarity to F strongly suggests a consanguineous origin and we interpret C to be a hematite echo of F, the latter most probably being carried by magnetite.

At five adjacent Lorrain sites, grayish red argillic siltstone and fine-grained sandstone possess two almost anti-parallel components, the reverse component E evidently being carried by fine-grained magnetite and the normal component B by hematite. The consistent structural attitude at the five sites did not permit fold tests. The Lorrain B and E in situ directions fail a reversal test [29]. However, the dip-corrected Lorrain B direction is very closely anti-parallel to the in situ Lorrain E direction and in that configuration the components pass a reversal test (rated B) at 95% confidence. The Lorrain B direction is close to that of Morris A [6].

Coleman sandstone A and Lorrain A components with unblocking temperatures near 690°C are well defined. Coleman-sandstone A is characterized by very shallow southwesterly directions (Fig. 4a), five sites near Elliot Lake (road cuts within 46°26.95–27.96′N, 82°37.67–40.06′W) giving a dip-corrected direction of $D = 216.3°$, $I = 5.5°$ ($\alpha_{95} = 6.7°$, $n = 30$). Lorrain A was identified at all but one site and is marked by shallow to very shallow northerly directions (Fig. 4b); 17 sites (road cuts within 46°20.51–22.00′N, 83°55.75–58.31′W) give a dip-corrected mean direction of $D = 5.4°$, $I = 5.5°$ ($\alpha_{95} = 5.9°$, $n = 75$). The consistent low (6°) northerly structural dips at the five Coleman-sandstone A sites did not permit a fold test, and unfavorable structural trends rendered inconclusive a fold test for Lorrain A.

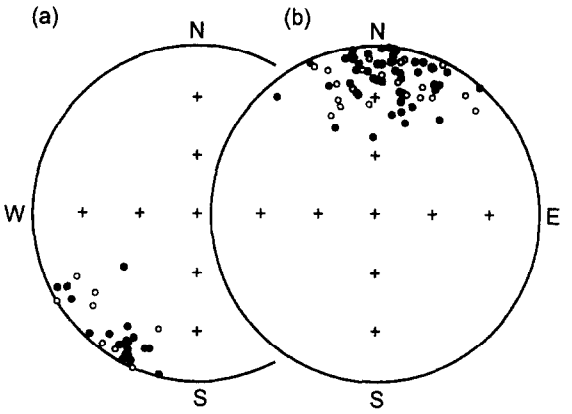


Fig. 4. Equal-area stereographic projections of dip-corrected specimen directions for (a) Coleman-sandstone A for five sites near Elliot Lake, and (b) Lorrain A for 17 sites near Desbarats. Closed symbols indicate down directions and open symbols up directions.

4.3. Interpretation of results

With the exception of Coleman-sandstone B, Coleman-diamictite B and Firstbrook C components, which are close to the present field direction and are probably viscous overprints given their low unblocking temperatures, all the directions given in Table 1 are plotted with their 95% confidence limits on an equal-area stereographic projection (Fig. 5). Directions previously determined for the Coleman and Firstbrook members [6,23–25] also are shown.

Coleman-diamictite A, although passing a fold test at 99% confidence, is viewed with the same caution afforded the Morris A component (see be-

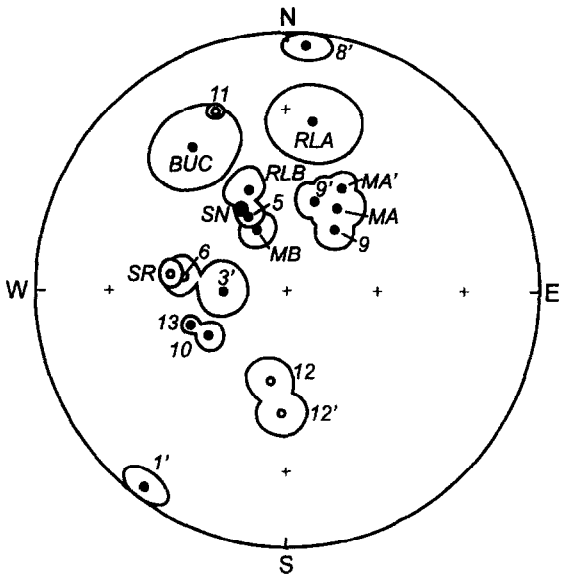


Fig. 5. Equal-area stereographic projections of directions with 95% confidence limits for components of the Gowganda and Lorrain formations. Closed symbols indicate down directions and open symbols up directions. Abbreviations are: *RLA*, Roy and Lapointe (Firstbrook) A [23]; *RLB*, Roy and Lapointe (Firstbrook) B [23]; *MA*, Morris (Coleman) A, in situ [6], *MA'*, Morris (Coleman) A, dip-corrected [6]; *MB*, Morris (Coleman) B [6]; *BUC*, Buchan et al. Coleman red bed [25]; *SN*, *SR*, Stupavsky et al. (Firstbrook) Normal, Reverse [24]. All numbers are keyed to Table 1. Prime indicates dip-corrected.

low), for similar reasons. Because of the Penokean age of folding, a fold test alone does not constrain the magnetization age to be close to the time of deposition. Moreover, the proximity of the Coleman-diamictite A dip-corrected direction to the mean direction for the 2167-Ma Biscotasing dykes ($D = 267.0°$, $I = 60.5°$; [30]) suggests that the ages of these components are similar and that folding post-dated Biscotasing time.

Firstbrook A and B, which yield negative fold tests, are interpreted as overprints acquired after folding or tilting. Firstbrook A is close to Roy and Lapointe B [23] and also to postfolding diagenetic or Penokean metamorphic directions [6,20,22]. We therefore conclude that Firstbrook A and Roy and Lapointe B were acquired during the Penokean Orogeny.

Directions for Lorrain C, D and F give negative fold tests or show substantial scatter on dip correction, and these components are interpreted as overprints acquired after folding. Lorrain D is close to the postfolding Dollyberry Lake basalts B [22].

Lorrain B and the reverse component Lorrain E occur in a local argillic facies that is comparable to the Firstbrook argillic siltstones for which only postfolding components have been identified. Because of the widespread occurrence of overprints carried by magnetite and hematite in Gowganda argillic rocks and doubts regarding the suitability of such facies in establishing primary directions, we hold reservations that Lorrain B and E were acquired near the time of deposition. The close proximity of the Lorrain dip-corrected B and in situ E directions may indicate that the B component was acquired immediately before folding and the E component just after folding. The geomagnetic field apparently changed polarity sometime between the acquisition of the two components. Hence Morris A [6] may date from the time of Lorrain B, immediately before folding, which would put the age of Morris A as Penokean. The difference between the Morris A/Lorrain B and Morris B [6] directions may reflect different timing associated with the pre-1850 Ma and post-1850 Ma phases of the Penokean Orogeny (see [16]).

The following points are relevant in assessing the origin of Coleman-sandstone A and Lorrain A components:

(1) The red bed sandstone facies and mineralogy of the Lorrain 'purple siltstone' member accord with those of 'Type A red beds' [31], for which the mineralogy at the time of deposition is fairly mature and remanence is an early chemical remanent magnetization (CRM).

(2) Many of the Coleman and Lorrain sandstone specimens contain vestiges of both normal and reverse polarity remanence (e.g. Fig. 3a,f), consistent with CRM acquired over a protracted interval of time and implying the identification of a dipole field axis and the sufficient averaging of secular variation. Therefore we have assigned unit weight to specimens in statistical analysis.

(3) Coleman-sandstone A and Lorrain A directions have not been identified in post-Huronian sedimentary or igneous rocks of the region, either as original or as overprint components, by us or numerous previous workers [10,25,32].

(4) The directions are, however, close to the dual polarity primary directions for the slightly older Matachewan dyke swarm in areas nearest our sites (D/I for Matachewan southern areas RA, RL and WL, respectively, $= 23.2°/17.9°$ 'normal', $204.1°/-11.6°$ 'reverse'; $19.2°/20.7°$, $209.2°/-6.0°$; $15.7°/24.9°$, $216.5°/-9.7°$ [33]).

(5) As noted above, a site of Buchan et al. [25] in Coleman red beds gave a shallow NNW-down direction that a positive conglomerate test suggested is not an overprint related to Nipissing intrusions. Applying the test of McFadden and Lowes [34] indicates that this Coleman direction is not significantly different from Lorrain A but is significantly different from Lorrain B. Roy and Lapointe's [23] Coleman A direction can belong either to Lorrain A or Lorrain B. Both in situ and dip-corrected directions for Lorrain D are significantly different from Lorrain A.

The above points suggest that Coleman-sandstone A and Lorrain A are not overprints related to later igneous events or tectonism, but more reasonably may be ascribed to early CRM acquired fairly soon after deposition.

Apparent polar wander may partly explain the 30° difference in mean declinations for Coleman-sandstone A and Lorrain A. However, substantial vertical axis rotations evidently occurred in the Canadian Shield and the Southern Province during the Proterozoic [35,36], and vertical axis rotation mostly in the central and northwestern Matachewan areas can account for the 40° spread of Matachewan declinations [33]. Hence the difference in Coleman-sandstone A and Lorrain A declinations for sampling areas on opposite sides of the Murray Fault Zone (Fig. 1), an important and long-lived Proterozoic fault system with ~2000 m strike-slip near its western limit [14,16], also may reflect some vertical axis rotation possibly related to movements along the Murray Fault Zone during early deformation in the south.

Van der Voo et al. [37] found that "inclination shallowing has not significantly contaminated the worldwide data-set of paleomagnetic results from redbeds". Because of the generally fine grain-size and good sorting of the sandstones analyzed here and the lack of evidence for detrital remanent magnetization, post-depositional compaction and related inclination shallowing are seen as unimportant also in the present study.

5. Discussion: paleolatitude of Huronian glaciation

Paleomagnetic poles for the main components identified by the present study (Table 1) are plotted on a global projection in Fig. 6 with paleomagnetic poles for Morris A and B [6], Roy and Lapointe B [23] and the Firstbrook directions of Stupavsky et al. [24]. Fig. 7 shows a global projection of paleomagnetic poles for Coleman-sandstone A and Lorrain A, and primary paleomagnetic poles for mafic dyke swarms of the southern Superior Province for 2.47–2.07 Ga [10]. In general, our data are consistent with Buchan et al.'s [10,30] revised Paleoproterozoic polar wander path for North America.

Viewing Morris A [6] as having been acquired near the time of deposition would indicate a moderate paleolatitude of $32.6° + 5.0°/-4.4°$, employing the geocentric axial dipole model for the geomagnetic field. However, the Morris A direction has not been found in the Gowganda Formation in the Elliot Lake and Cobalt areas by previous or later workers [18,20,22–24] or by us, and because of doubts regarding the suitability of Huronian argillic facies in establishing primary directions and the weak con-

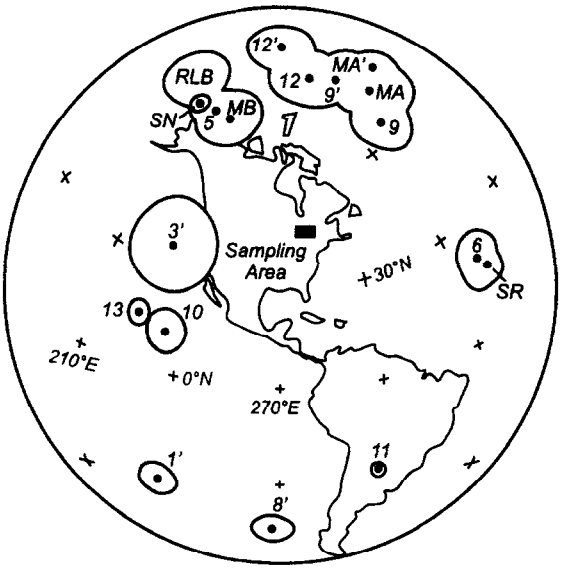


Fig. 6. Paleomagnetic poles for components of the Gowganda and Lorrain formations (polarities not implied). Abbreviations are as for Fig. 5, and all numbers are keyed to Table 1. Prime indicates dip-corrected.

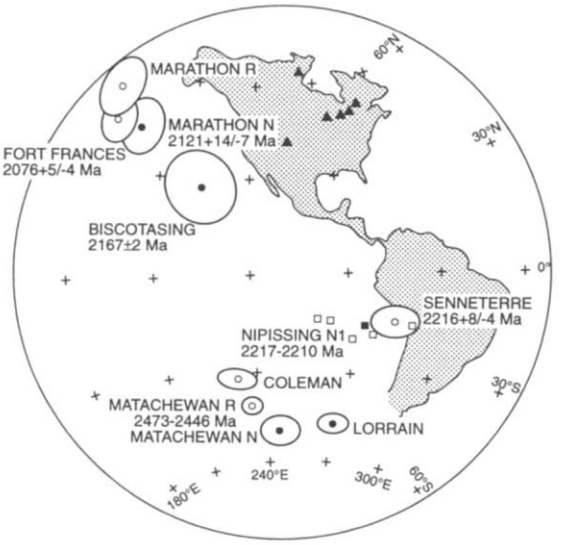


Fig. 7. Primary paleomagnetic poles for mafic dyke swarms of the southern Superior Province for 2.47–2.07 Ga (after [10]) and paleomagnetic poles for Coleman-sandstone A and Lorrain A components in the Southern Province (this study). Paleoproterozoic glaciogenic rocks in North America [38] are shown as closed triangles.

straints imposed by Morris's [6] fold test we hold reservations that Morris A was acquired near the time of deposition. Our Lorrain results suggest rather that Morris A may have been acquired just prior to Penokean folding.

On the other hand, ascribing Coleman-sandstone A and Lorrain A to CRM acquired soon after deposition implies equatorial paleolatitudes of $2.8° \pm 3.4°$ and $2.8° \pm 3.0°$, respectively. Viewed as such, the Matachewan, Coleman-sandstone A and Lorrain A poles may constrain Huronian paleolatitudes. Field and paleomagnetic observations are consistent with there having been only one field reversal (reverse → normal) during Matachewan time, with variation in normal and reverse mean inclinations possibly reflecting apparent polar wander during the evolution of the swarm [33]. The Matachewan normal direction has a mean inclination of 22.1° and the reverse direction a mean inclination of −15.7° (from data in [33]), indicating intrusion of the dyke swarm at paleolatitudes of about 11.5° to 8.0° prior to Huronian deposition. However, Matachewan southern areas RA, RL and WL nearest our sites have undergone minimal structural distortion since dyke intrusion [33] and may thus provide the most reliable estimates of undisturbed Matachewan directions; normal and reverse mean inclinations for those areas are 22.0° and −8.6°, indicating paleolatitudes of 11.4° to 4.3°. The Matachewan, Coleman-sandstone A and Lorrain A results therefore suggest that virtually all the Huronian Supergroup, including the glaciogenic Ramsay Lake, Bruce and Gowganda formations, accumulated within 11° and possibly within 4° of the paleoequator.

In conclusion, components carried by magnetite and hematite in Huronian *argillic* rocks, including Morris A [6], are suspect as original directions. Hence, the interpretation of low paleolatitudes for Huronian glaciation, based on high quality data for dyke swarms in the southern Superior Province and apparent early CRM in red sandstones from the Cobalt Group, appears to us to be preferred. The latter interpretation would be consistent with findings inferred from volcanics that Paleoproterozoic glaciation in South Africa occurred in low paleolatitudes [5]. While available paleomagnetic data rule out high paleolatitudes for Huronian glaciation, our findings alone cannot discriminate among possible explanations of low paleolatitude glaciation during the Precambrian [39–42]. Resolution of the enigma of low paleolatitude glaciation will require concentrated paleomagnetic and paleoenvironmental studies of Precambrian glaciogenic successions throughout the world together with assessment of the configuration of the Precambrian geomagnetic field.

Acknowledgements

We thank Grant Young for much assistance during the initial sampling in 1995, Dave Clark for commenting on the manuscript, Ross Both for assistance with petrography, and Dave Evans, Joe Kirschvink and Joe Meert for constructive reviews. Williams is supported by the Australian Research Council. *[RV]*

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